

RAHUL KP

Software Development Engineer | Backend & AI/ML Kerala, India · +91 79091 52002 · rahulkurup@gmail.com LinkedIn · GitHub · Portfolio · Hugging Face

SUMMARY

Software engineer with an **M.Sc. in Computer Science** and hands-on experience building backend systems, APIs, distributed application architectures, and machine-learning systems. Strong foundation in **Python, C/C++, data structures & algorithms, SQL, FastAPI, PostgreSQL, Redis, Docker, and system design**, complemented by deep-learning and recommendation-system experience. Interested in building **reliable, high-performance backend systems, payment/business logic, API integrations, and scalable infrastructure**.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL **Backend:** FastAPI, REST APIs, Pydantic, SQLAlchemy, PostgreSQL, Redis **Engineering:** Data Structures & Algorithms, OOP, System Design, Authentication, API Integration, Testing **Frontend:** Next.js, React, TypeScript, Tailwind CSS **DevOps:** Docker, Docker Compose, Nginx, GitHub Actions, Kubernetes, Prometheus, Grafana **AI/ML:** NumPy, PyTorch, Scikit-learn, Neural Networks, Deep Learning, NLP, LLMs, RAG, Recommender Systems **Foundations:** Linear Algebra, Probability, Statistics, Machine Learning Mathematics, Functional/Declarative Programming Concepts

SELECTED PROJECTS

NCF Recommender System — End-to-End Recommendation Platform

Python · NumPy · PyTorch · FastAPI · Next.js · PostgreSQL · Docker · Nginx

- Built a **production-oriented hybrid recommendation system** combining Neural Collaborative Filtering, content-based filtering, and popularity-based recommendations.
- Implemented **NCF from first principles with NumPy**, then benchmarked against an optimized PyTorch implementation with MPS/GPU support.
- Designed a **microservice architecture** separating frontend, API/backend, recommendation engine, and model serving.
- Addressed the **cold-start problem** using hybrid recommendation strategies and implemented movie search/ranking and interaction tracking.
- Achieved **0.6293 Hit@10** and **0.3541 NDCG@10** with the PyTorch model; documented experiments, architecture, and evaluation methodology.

RankScript — Competitive Learning Platform

FastAPI · Next.js · PostgreSQL · Redis · Docker · JWT

- Engineered a full-stack learning platform with **Student, Mentor, and Admin** roles and role-based access control.
- Developed REST APIs using a layered **Route → Service → Model** architecture with SQLAlchemy and Pydantic.
- Implemented course management, quizzes, assignments, authentication, analytics, and competitive district/state/national leaderboards.
- Built a weighted ranking system combining **quiz performance, assignments, course completion, and learning streaks**.
- Used **PostgreSQL** for persistent data, **Redis** for caching, **Docker** for containerization, and automated CI/CD.

ANN Foundation — Neural Network & Autograd Engine

Pure Python · Automatic Differentiation · Neural Networks · Testing

- Implemented a neural-network learning pipeline **from mathematical first principles**, without PyTorch or TensorFlow.
- Built reverse-mode **automatic differentiation**, dynamic computation graphs, gradient accumulation, and reverse topological backpropagation.
- Implemented **tanh, ReLU, and sigmoid** activations plus a composable **Neuron → Layer → MLP** architecture.
- Created **19 gradient-verification tests** comparing analytical gradients against central-difference numerical differentiation.
- Added GitHub Actions CI/CD and deployed an interactive Gradio demonstration on Hugging Face Spaces.

RESEARCH

Neural Collaborative Filtering Recommender System Research/preprint work covering NCF implementation from first principles, NumPy vs. PyTorch benchmarking, hybrid recommendation architecture, evaluation metrics, and reproducible experiments.

EDUCATION

M.Sc. Computer Science — University of Calicut | **2024–2026 B.Sc. Computer Science** — University of Calicut | **2020–2023 HSE Computer Science** — Kerala Board | **2017–2019**

CERTIFICATIONS

- **IBM Generative AI Engineering Professional Certificate**
- Oracle Cloud Infrastructure — Generative AI Professional
- Google — Accelerate Your Job Search with AI
- Additional industry certifications and courses in AI, software engineering, communication, and professional development

LINKS

Portfolio: <https://rahulkp-ai.github.io/> **GitHub:** <https://github.com/rahulkp-ai> **LinkedIn:** <https://linkedin.com/in/rahulkp-ai> **Hugging Face:** <https://huggingface.co/rahulkp-ai> **ORCID:** <https://orcid.org/0009-0009-3403-6670>